Two Chemical Windows on Dengue: Non-Overlapping Raman and SERS Signatures under a Common Chemometric Protocol

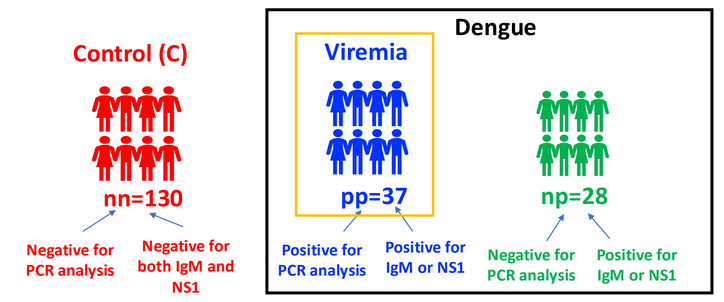
Leonardo C. Pacheco-Londoño[1]*, Aileen Y. Chang[2] , Juan David Álzate[2], María José Diaz[2], Samuel P. Hernández-Rivera[3], Nataly Galan-Freyle[1], Lisandro Pacheco-Lugo[1], Martha Guerra-Simanca[1], Leonardo Arrieta-Rangel[1], Susana Ospina-Maldonado[1] and Antonio Acosta-Hoyo[1]

1 Life Science Research Center (CICV), Universidad Simón Bolívar, Barranquilla, Colombia. 2 George Washington University, Washington DC, USA. 3 Center for Chemical Sensors (CCS), Chemical Imaging and Surface Analysis Center (CISAC), Department of Chemistry, University of Puerto Rico-Mayagüez, Mayagüez, PR 00681, USA *leonardo.*pacheco@unisimon.edu.co*

1 · BACKGROUND

- Dengue and chikungunya overlap clinically in the first days of illness, exactly when confirmatory testing (RT-PCR, NS1, IgM) is least sensitive or unavailable at first contact.
- Label-free Raman and SERS of serum read a systemic biochemical response from microliters of a minimally processed sample.
- **Comparisons between Raman approaches are only interpretable when the chemometric protocol is held fixed.**

2 · THREE DATASETS, ONE PROTOCOL

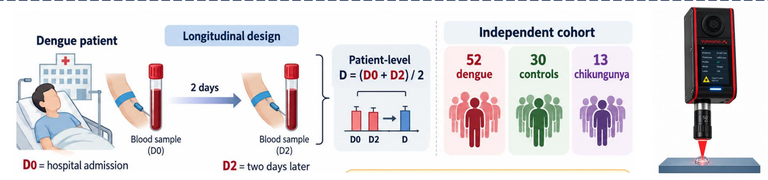


Conventional Raman · 785 nm

5 µL serum droplets, portable 785 nm system (Wasatch Photonics 785L). 195 participants: 65 dengue (37 RT-PCR+, 28 RT-PCR−/serology+) and 130 healthy controls.

AuNP-SERS · 785 nm

Centrifuged serum supernatant (5 µL serum droplets) + gold nanoparticles (5 µL serum droplets). 58 patients of the SAME first cohort: 35 RT-PCR-positive, 23 RT-PCR-negative / serology-positive.



AgNP-SERS · 830 nm

Serum/plasma + AgNP, independent cohort, longitudinal: D0 at hospital admission, D2 two days later; patient-level D = (D0 + D2)/2. 52 dengue, 30 controls, 13 chikungunya — dengue vs chikungunya also resolved.

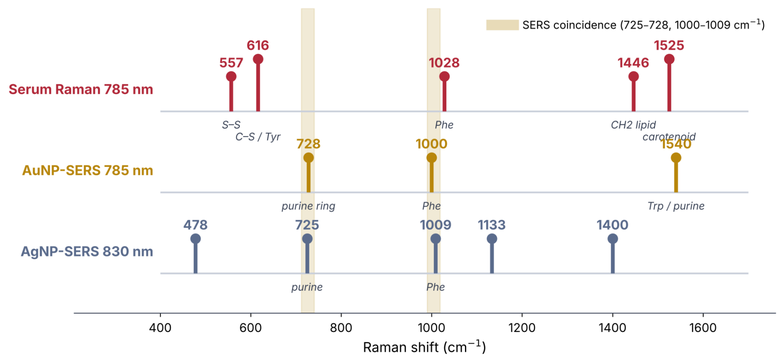
3 · ONE PATIENT-LEVEL PROTOCOL

same chemometrics for all three platforms

Serum (1 obs / patient) → 3 platforms (785 · AuNP · 830 AgNP) → Preprocessing (SG + AsLS) → PLS-DA (100 re-splits) → Se · Sp · AUC + VIP bands

No pseudoreplication · Savitzky–Golay + AsLS (λ = 10^4, p = 0.001) · LV selection by CV inside the training block only · external test + 100 stratified 75/25 re-splits · discriminant bands by VIP

4 · ALMOST DISJOINT DISCRIMINANT BANDS



VIP bands per platform. Raman 785 reads disulfide, aromatic, lipid and carotenoid modes; Ag- and Au-SERS are dominated by purine and aromatic amino-acid bands, coinciding at 725–728 and 1000–1009 cm^{-1} and sharing only 616–640 cm^{-1} with Raman.

5 · FIVE MODELS, COMPARABLE PERFORMANCE

Model · contrast	Sensitivity	Specificity	AUC
Raman 785 · M1-DS dengue screening: dengue vs control	0.81 ± 0.09	0.91 ± 0.04	0.93 ± 0.03
Raman 785 · M2-VS viremia stratification: RT-PCR+ vs RT-PCR−/control	0.84 ± 0.09	0.96 ± 0.03	0.995 ± 0.008
AuNP-SERS 785 RT-PCR+ vs RT-PCR− (same sera)	0.87 ± 0.12	0.88 ± 0.15	0.96 ± 0.06
AgNP-SERS 830 dengue vs control	0.854 ± 0.103	0.88 ± 0.12	0.93 ± 0.06
AgNP-SERS 830 dengue vs control + CHIK	0.831 ± 0.101	0.88 ± 0.09	0.93 ± 0.05
AgNP-SERS 830 dengue vs CHIK	0.97 ± 0.05	0.86 ± 0.20	0.98 ± 0.3

Mean ± SD over 100 stratified 75/25 re-splits (patient level). Raman 785 accuracies: 0.88 ± 0.04 (M1-DS) and 0.93 ± 0.03 (M2-VS). AUC computed here over the same 100 re-splits (PLS-DA, 4/7 LVs, AsLS λ = 100). AuNP classes were acquired in separate sessions and its controls followed a different preparation protocol, so only the viremia contrast is reported (permutation-null balanced accuracy 0.48). All chikungunya test patients were assigned to the non-dengue class.

Se 0.81–0.87 · Sp 0.88–0.96

AUC 0.93–0.99

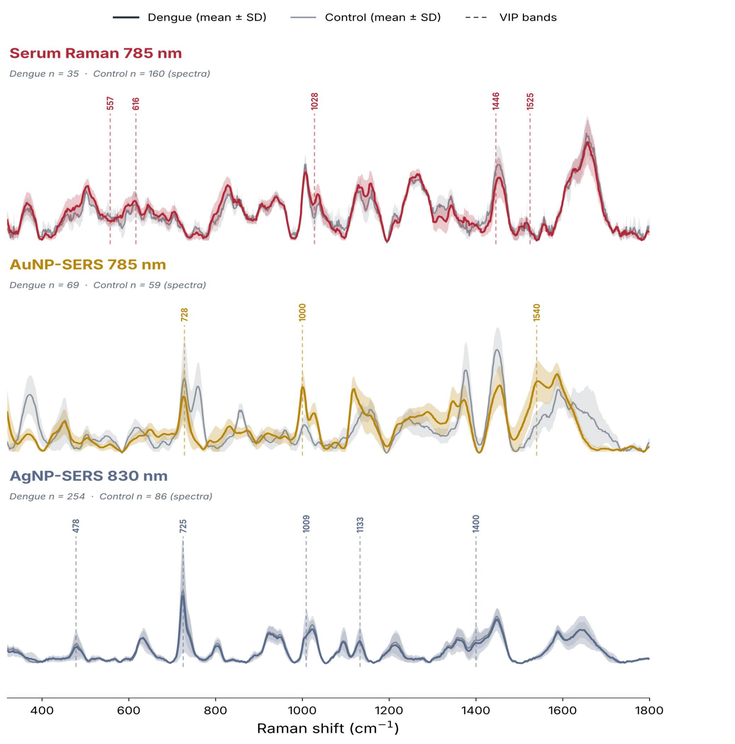
PLATFORM HIGHLIGHTS

Raman 785 · single-band ratios 1028/616 and 1525/640 alone reach AUC 0.83–0.93, defining a three-zone rule: healthy → viremic → declining viremia. r = 0.980.

AuNP-SERS 785 · without AuNP no serum fraction separates serology (balanced acc. 0.55–0.63, ~chance): the diagnostic signal requires the nanoparticles. 1540 and 1000 cm^{-1} rise in PCR+, 728 cm^{-1} in PCR−.

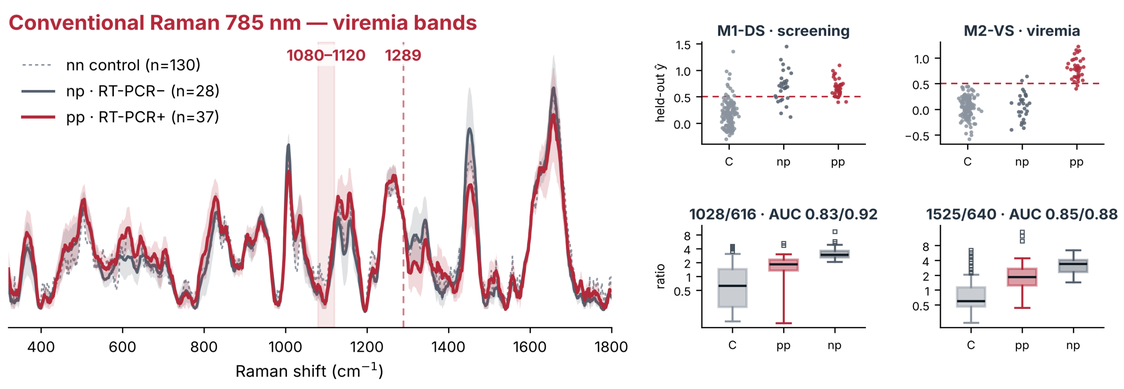
AgNP-SERS 830 · the D0 → D2 change is a global SERS-intensity gain; only 840 cm^{-1} survives vector normalization. Adding chikungunya leaves the AUC unchanged (0.928 vs 0.926).

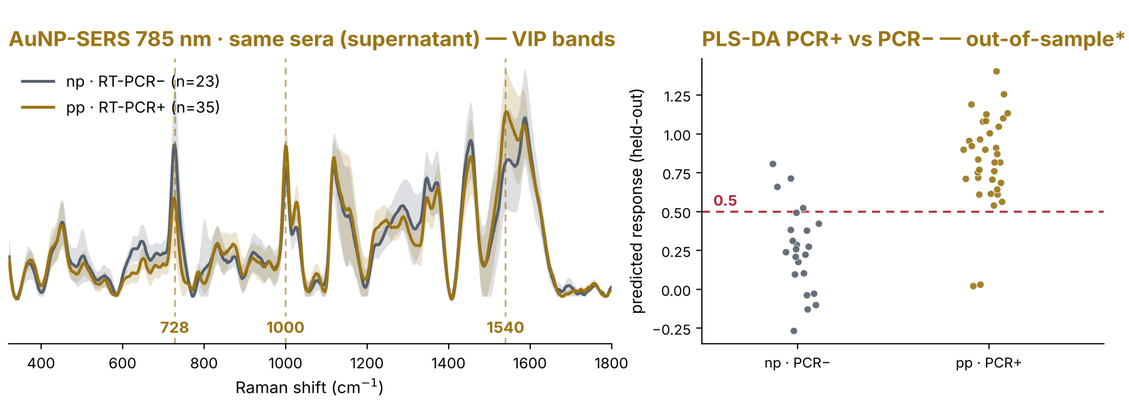
6 · MEAN SPECTRA



Mean ± SD, dengue vs control (AsLS + L2, 320–1800 cm^{-1}); dashed guides = VIP bands.

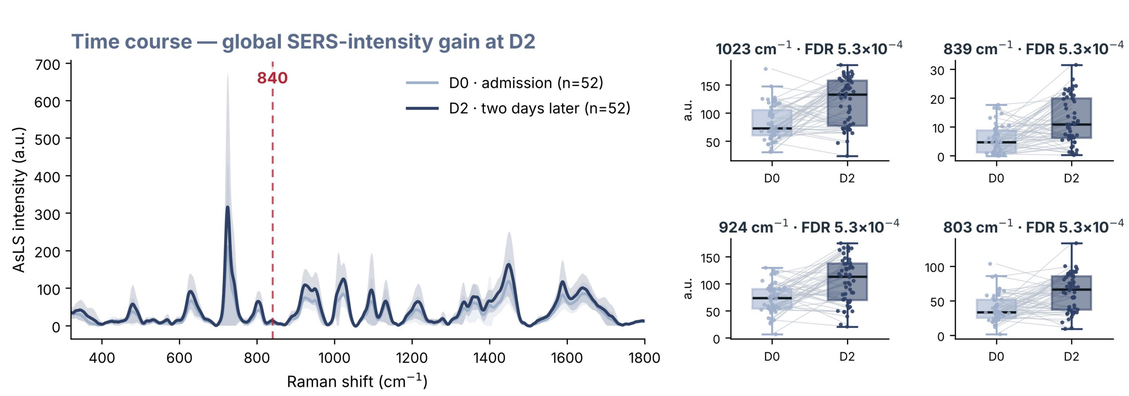
785 nm — SAME PATIENTS, TWO WINDOWS

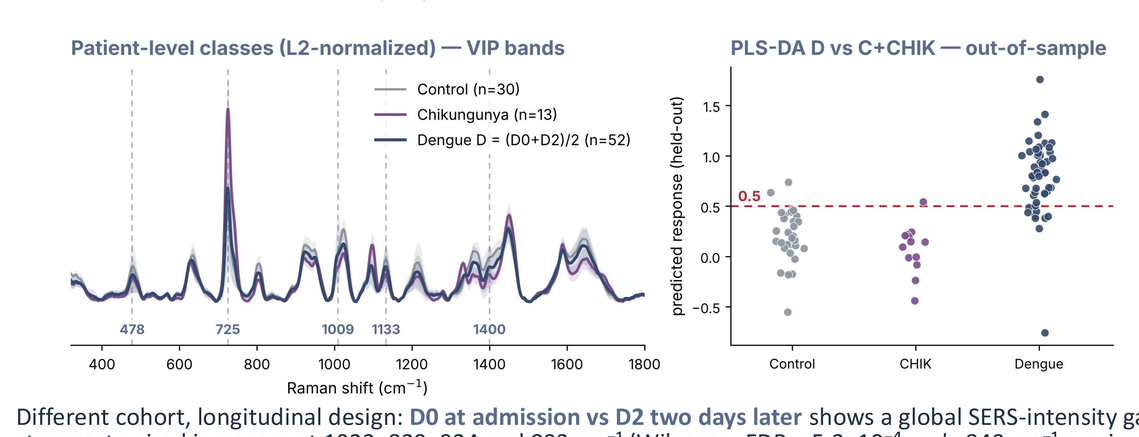




Same first cohort: 58 of the 65 dengue patients were also measured by AuNP-SERS. **Conventional Raman resolves viremia** through amide III 1289 and 1080–1120 cm^{-1} (Se 0.84, Sp 0.96); on the same sera, **AuNP-SERS reads purine/aromatic bands instead** (Se 0.872, Sp 0.877, AUC 0.956*). The held-out PLS-DA responses (right) separate the classes on both platforms; the divergence belongs to the platform — not the cohort.

AgNP-SERS 830 nm — INDEPENDENT COHORT





Different cohort, longitudinal design: **D0 at admission vs D2 two days later** shows a global SERS-intensity gain — strongest paired increases at 1023, 839, 924 and 803 cm^{-1} (Wilcoxon, FDR = 5.3×10^{-4}; only 840 cm^{-1} survives L2 normalization). The held-out PLS-DA response separates **dengue from controls AND chikungunya** (AUC 0.926 ± 0.065 and 0.928 ± 0.051); CHIK patients fall on the non-dengue side.

7 · CONCLUSIONS

1. Five PLS-DA models, three platforms: Se 0.81–0.87, Sp 0.88–0.96, AUC 0.93–0.99 under one patient-level protocol.
2. Almost disjoint bands: Raman reads disulfide, aromatic, lipid and carotenoid modes; Ag- and Au-SERS are purine-dominated.
3. Raman and SERS act as two complementary chemical windows on the dengue serum phenotype.
4. Prospective validation, single-sample models and same-protocol controls are required before any diagnostic claim.

FUNDING

Minciencias — Sistema General de Regalías (BPIN 2024000100078).

QR (paper / contact)